CS283 Syllabus
Systems Programming
Summer 2026

Instructor Information:
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Catalog Description: This course introduces computer systems, including interaction of hardware and software through the operating system, from the programmer's perspective. Three fundamental abstractions are emphasized: processes, virtual memory, and files. These abstractions provide programmers a common interface to a wide variety of hardware devices. Topics covered include linking, system level I/O, concurrent programming, and network programming.

Text: Rather than use a conventional textbook, I will make available on Blackboard a number of key papers and man pages that will be the important references for you this term.

Objectives: Every piece of code you ever write runs in some context and some environment, most often an operating system. The primary focus of this course is to understand the relationship between applications and the operating system. The second objective is to learn how to use the services that the operating system provides for application programmers. Learning how to use them does not mean learning how to search for examples to copy or how to prompt an LLM to write the code. It means learning how to read the man pages and other documentation and writing the code from scratch. Along the way, we're going to learn C and learn how to use it the right way. We will study and understand the tools that make up the software development ecosystem. We will develop a thorough understanding of the application side of the application/OS interface. Finally, we'll pick up some experience in thread and socket programming.

Topics:
- The C Programming Language
- Editors, compilers, and linkers
- System calls and basic libraries for processes, memory, and files
- The shell and other core systems programs
- POSIX threads
- Network programming with sockets

Grading: Because of changes over the past few years in how students approach their work, the grading in this class will be more dependent on face-to-face exams than in the past. The exact number and weighting is TBD, but I'm expecting about seven or eight programming assignments and three exams. The exams will probably be worth about 65% of the course grade and the projects the remaining 35%. Some of the exams will test your understanding and ability by presenting you with man pages for one or more system or library calls and expecting you to write some code with them on paper and without any electronic assistance.

Project Submissions: You should submit projects electronically via Blackboard. Submit each file of a project individually, rather than using any form of archiving, especially zip. Do NOT include any spaces in file names. Textual submissions are subject to some constraints. I do not have the time, money, disk space or inclination to support every commercial, proprietary, closed, undocumented word processor format out

there. For that reason, I do not accept submissions in any of them. You must submit your assignments in an open well-documented format. One option is plain ASCII text. If you need to express equations or some other material that you feel you can't express well in plain ASCII, then I encourage you to use a typesetting language such as troff or TEX. If you are using a word processor, then save the file as PDF and submit that.

Development Environment: Because the primary focus of this course is on serious programming that interacts closely with the operating system, I expect all work to be done on tux. Furthermore, an important part of the experience is to use the tools on tux, including the editors. You are expected to choose from editors such as ed, vi, or emacs. If you want to be adventurous, you can try some of the later editors that came out of Bell Labs like sam or acme. Editors/IDEs such as vscode or eclipse are NOT to be used.

Academic Integrity: All written solutions and code must be the result of individual effort. If you have any questions about what is permissible, ask before referring to any resources not provided by the instructor. Please note that code does not have to be an exact copy to be a violation of academic integrity. Working on code together and typing it independently is not individual effort. Although copying elements of programming "examples" to create new programs is commonly done, it is not an acceptable technique in this course.

Unapproved collaboration or use of code from unapproved sources are violations of academic integrity. Additionally, any posting of questions, assignments, or exams from this course to any external discussion sites will be considered a violation of academic integrity. This includes, but is not limited to, Chegg, CourseHero, Koofers, Slader, stackoverflow, and Reddit.

Furthermore, use of any generative AI system, such as ChatGPT, Claud, etc. is strictly forbidden in this class. This includes things like CoPilot in an IDE and even when you write e-mails to me. The sanction for academic integrity violations can range from a zero grade on the assignment to summary failure for the course at the discretion of the instructor.

Incomplete Policy: Recent changes to university incomplete policy have made incompletes essentialy unworkable. Consequently, I do not intend to do incomplete any longer. If you have or anticipate difficulty in getting all of the assignments completed, you need to talk to me directly as early as possible.

Academic Policies: This course follows university, college, and department policies, including but not limited to:

- Academic Integrity, Plagiarism, Dishonesty and Cheating Policy:
  `http://www.drexel.edu/provost/policies/academic_dishonesty.asp`
- Student Life Honesty Policy from Judicial Affairs:
  `http://www.drexel.edu/provost/policies/academic-integrity`
- Students with Disability Statement:
  `http://drexel.edu/oed/disabilityResources/students/`
- Course Add/Drop Policy:
  `http://www.drexel.edu/provost/policies/course-add-drop`
- Course Withdrawal Policy:
  `http://drexel.edu/provost/policies/course-withdrawal`
- Department Academic Integrity Policy:
  `http://drexel.edu/cci/resources/current-students/undergraduate/policies/cs-academic-integrity/`
- Drexel Student Learning Priorities:
  `http://drexel.edu/provost/assessment/outcomes/dslp/`
- Office of Disability Resources:
  `http://www.drexel.edu/ods/student_reg.html`

The instructor(s) may, at his/her/their discretion, change any part of the course before or during the term, including assignments, grade breakdowns, due dates, and schedule. Such changes will be communicated to

students via the course web site. This web site should be checked regularly and frequently for such changes and announcements.

Students requesting accommodations due to a disability at Drexel University need to request a current Accommodations Verification Letter (AVL) in the ClockWork database before accommodations can be made. These requests are received by Disability Resources (DR), who then issues the AVL to the appropriate contacts. For additional information, visit the DR website at

`http://drexel.edu/oed/disabilityResources/overview/`

or contact DR for more information by phone at 215.895.1401, or by email at `disability@drexel.edu`.